ALEJANDRO FONTAL

RESEARCH DATA SCIENTIST

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WORKING EXPERIENCE

Climate & Health Program @ ISGlobal

Post-doctoral Researcher

Oct. 2024 – Present Barcelona, ES

- Serving as the group's software and data engineering lead. Development of packages, webapps and dashboards.
- Building reproducible pipelines for long-read aerobiome metagenomics and linking results to health outcomes.
- Repurposing a pollen detector for bacterial discrimination, combining laser-induced fluorescence, light scattering, and supervised ML for near-real-time identification.

PhD. Fellow - Early Stage Researcher (ESR)

Oct. 2019 – Oct. 2024

Marie Skłodowska-Curie Actions ESR within *HELICAL ITN*.

- Developed time-series methods to quantify environmental signals in epidemics, including transient and lagged associations, multi-scale correlations, and phase-shift analyses.
- Integrated nationwide case data with climate reanalysis, remote sensing, GIS layers, and air-mass back-trajectories to model drivers of Kawasaki Disease, COVID-19, influenza.
- Coordinated in-situ sampling campaigns in Japan to characterize biological and chemical fractions of air-masses.

Docker | Nextflow | Dash | quarto | HPC | GIS | GH Actions

Protein Engineering @ DuPont Industrial Biosciences

Data Scientist

Sep. 2018 – Aug. 2019 Leiden, NL

- Pioneered R&D efforts to integrate data-driven solutions to the protein engineering workflow, including but not limited to ML.
- Built and maintained reproducible bioinformatics and data pipelines.
- Partnered with wet-lab teams to design experiments and close the model-experiment loop, accelerating the automation of workflows.

Graduate Trainee in Data Science

Mar. 2018 – Sep. 2018

MSc Internship project: **ML for Empirical Protein Design**

- Built, trained and benchmarked deep learning models to predict enzyme performance from sequence and structure.

Pytorch | Tensorflow | GANs | CNNs | LSTMs | Gitlab CI/CD

Educational Staff Development @ Wageningen University

Assistant in MOOCs Development

Sep. 2016 – Dec. 2017 Wageningen, NL

- Developed assignments and content for 10+ edX MOOCs.
- Provided technical support.

EDUCATION

PhD. in Biotechnology

Universitat de Barcelona

2019-2024 Barcelona, ES

Specialized in Data science & Epidemiology.

MSc. in Bioinformatics

Wageningen University & Research

2016-2018 Wageningen, NL

Data Science minor, *cum laude* distinction. Thesis on explainable deep learning models for protein subcellular location prediction.

BSc. in Biotechnology

Universitat de Barcelona

2011-2015 Barcelona, ES

Thesis at Bioinformatics unit @ VHIR on metaanalysis of transcriptomics tools.

Certified English C2 level.

DEVELOPMENT SKILLS

Python: High proficiency; 9+ years. Main language.

- Package development and testing.
- End-to-end data science: wrangling, viz, modelling, and webapp deployment (*pandas*, *polars*, *statsmodels*, *scikit-learn*, *tensorflow*, *pytorch*, *plotnine*, *dash*).

R language: Experienced with tidyverse (*ggplot2*, *dplyr*) and Bioconductor; frequent R–Python polyglot workflows.

Shell and CLI: Daily Linux user; HPC workflows with SLURM and Nextflow.

Git: Collaborative development; CI/CD with GitHub Actions and GitLab CI/CD.

Web: Basics of HTML/CSS/JS; build small tools and dashboards, simple REST endpoints, and deploy static sites.

Systems and Networking: Maintain small Linux servers; run Docker stacks behind nginx/Traefik with Let's Encrypt; manage SSH/users; automate with `systemd/cron`; operate a WireGuard VPN.